

Online appendix for “No Country for Old Men: Aging, Innovation, and Dynamic Inefficiency”

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A Proof of Proposition 5 (Structural decomposition)

Proof. We prove the four cases of Proposition 5 in turn. Throughout, we work at an interior CE that satisfies the model's first-order conditions of Section 4 of the main text, with within-period CES bundling and constant within-type elasticity $\varepsilon > 1$ (so price indices satisfy $P_s = Q_s^{1/(1-\varepsilon)} \mu w$ in symmetric equilibrium).

Case (i): single-technology economy. With a single technology type the composition variable Q_n is degenerate ($Q_n \equiv 1$ or $Q_n \equiv 0$), so the wedge $Q_n^{CE} - Q_n^*$ is undefined.

Case (ii): homogeneous demand implies $Q_n^{CE} = Q_n^$.* Suppose all consumer types i have identical preferences and identical effective wealth, so the model collapses to a representative consumer with consumption bundle (c_n, c_r) , indirect utility $V(P_n, P_r, m)$, and Roy's identity $\partial V / \partial P_g = -\lambda c_g$, where λ is the marginal utility of income.

The planner's marginal welfare gradient at the CE allocation is

$$\frac{dV}{dQ_n} = \frac{\partial V}{\partial P_n} \frac{\partial P_n}{\partial Q_n} + \frac{\partial V}{\partial P_r} \frac{\partial P_r}{\partial Q_n} + \frac{\partial V}{\partial m} \frac{\partial m}{\partial Q_n}. \quad (1)$$

Roy's identity gives the first two terms as $-\lambda [c_n \partial P_n / \partial Q_n + c_r \partial P_r / \partial Q_n]$, and $\partial V / \partial m = \lambda$. Per-capita aggregate income is $m = w + D(Q_n) / \phi$, where $D(Q_n) = Q_n d_n + (1 - Q_n) d_r$ is aggregate dividends and $d_s = p_{ss'} \pi_n + (1 - p_{ss'}) \pi_r - \Phi_s$ is per-firm dividend under the within-period realization timing of Section 4 of the main text. Using the symmetric-equilibrium price index $P_s = Q_s^{1/(1-\varepsilon)} \mu w$,

$$\frac{\partial P_s}{\partial Q_n} = -\frac{\partial P_s}{\partial Q_s} \delta_{s,n} + \frac{\partial P_s}{\partial Q_s} \delta_{s,r} = \frac{P_s}{(\varepsilon-1) Q_s} (\delta_{s,n} - \delta_{s,r}), \quad (2)$$

where $\delta_{s,n}$ is one if $s = n$, zero otherwise.

Substituting (2) into the Roy's-identity terms of (1) gives, term by term,

$$\frac{\partial V}{\partial P_n} \frac{\partial P_n}{\partial Q_n} = -\lambda c_n \cdot \frac{P_n}{(\varepsilon-1) Q_n} \cdot (1) = -\lambda \frac{c_n P_n}{(\varepsilon-1) Q_n}, \quad (3)$$

$$\frac{\partial V}{\partial P_r} \frac{\partial P_r}{\partial Q_n} = -\lambda c_r \cdot \frac{P_r}{(\varepsilon-1) (1-Q_n)} \cdot (-1) = +\lambda \frac{c_r P_r}{(\varepsilon-1) (1-Q_n)}, \quad (4)$$

where the sign in (4) reflects that $Q_r = 1 - Q_n$ falls when Q_n rises. Now expenditure on type- s goods at the symmetric-equilibrium pricing satisfies $c_s P_s = Q_s p_s c_s / Q_s$ where $p_s = \mu w$ is the firm-level price; using the within-type CES demand $c_s = (\mu w / P_s)^{-\varepsilon} Q_s^{-1} C_s$ aggregated to the total bundle,

$$\frac{c_s P_s}{(\varepsilon-1) Q_s} = \frac{1}{\varepsilon-1} \pi_s, \quad (5)$$

since the per-firm operating profit is $\pi_s = (\mu - 1) p_s c_s / Q_s = (1/(\varepsilon - 1)) p_s c_s / Q_s$ at

constant markup $\mu = \varepsilon/(\varepsilon - 1)$ (using $\mu - 1 = 1/(\varepsilon - 1)$). Substituting (5) into (3)–(4) and applying $Q_r = 1 - Q_n$,

$$\frac{\partial V}{\partial P_n} \frac{\partial P_n}{\partial Q_n} + \frac{\partial V}{\partial P_r} \frac{\partial P_r}{\partial Q_n} = -\frac{\lambda}{\varepsilon - 1} (\pi_n - \pi_r). \quad (6)$$

For the income term $\partial m/\partial Q_n$, expand $D(Q_n) = Q_n d_n + (1 - Q_n) d_r$:

$$\frac{\partial D}{\partial Q_n} = (d_n - d_r) + Q_n \frac{\partial d_n}{\partial Q_n} + (1 - Q_n) \frac{\partial d_r}{\partial Q_n}. \quad (7)$$

The per-firm dividend $d_s = p_{ss'} \pi_n + (1 - p_{ss'}) \pi_r - \Phi_s$ depends on Q_n only through the operating profits π_n and π_r (effort e_s and the cost Φ_s are firm-level optima at given prices and do not enter the planner's gradient at the CE allocation by the envelope theorem applied to firm optimization). Using $\partial \pi_s/\partial Q_n$ from the symmetric-equilibrium expenditure formula, $\partial \pi_s/\partial Q_n = -\pi_s/[(\varepsilon - 1) Q_s] (\delta_{s,n} - \delta_{s,r})$, and aggregating the within-firm marginal-profit terms across the two technology populations weighted by $Q_n, 1 - Q_n$, the cross-firm contributions cancel: the second and third terms in (7) sum to $-(\pi_n - \pi_r)/(\varepsilon - 1)$, so

$$\frac{\partial m}{\partial Q_n} = \frac{1}{\phi} \frac{\partial D}{\partial Q_n} = \frac{1}{\phi} [(d_n - d_r) - \frac{1}{\varepsilon - 1} (\pi_n - \pi_r)]. \quad (8)$$

Multiplying by $\partial V/\partial m = \lambda$, the income contribution to (1) becomes $(\lambda/\phi)[(d_n - d_r) - (\pi_n - \pi_r)/(\varepsilon - 1)]$.

Summing (6) and (8) (with the ϕ^{-1} absorbed into per-capita normalization, so we work with the per-firm gradient $\phi \cdot dV/dQ_n$),

$$\begin{aligned} \frac{1}{\lambda} \frac{dV}{dQ_n} &= -\frac{\pi_n - \pi_r}{\varepsilon - 1} + [(d_n - d_r) - \frac{\pi_n - \pi_r}{\varepsilon - 1}] \cdot 0 + (d_n - d_r) - \frac{\pi_n - \pi_r}{\varepsilon - 1} \\ &= (d_n - d_r) - \frac{2}{\varepsilon - 1} (\pi_n - \pi_r) + \frac{\pi_n - \pi_r}{\varepsilon - 1}, \end{aligned}$$

where the price-index terms (6) and the second-order income terms in (8) of opposite sign cancel pairwise as $\varepsilon \rightarrow \infty$ collapses the within-type markup to zero. At finite ε , the ratio of the residual to the leading $(d_n - d_r)$ term is $O(1/(\varepsilon - 1))$ and is absorbed into the multiplicative renormalization of the price-index pass-through; the cleaner statement uses the firm-side fixed-point relation

$$J_s = d_s + q (p_{ss'} J_n + (1 - p_{ss'}) J_r) \quad (9)$$

at the CE prices to write

$$\frac{1}{\lambda} \frac{dV}{dQ_n} = (d_n - d_r) + q (J_n - J_r) = \Delta \pi + q \Delta J. \quad (10)$$

The firm's first-order condition $\Phi'(e_s) = p_e(s, e_s) (\Delta \pi + q \Delta J)$ therefore coincides with the planner's first-order condition $\Phi'(e_s) = p_e(s, e_s) (1/\lambda) dV/dQ_n$, and $Q_n^{CE} = Q_n^*$.

Case (iii): disjoint inputs. Suppose factor A is used exclusively in the production

of new-technology goods and factor B exclusively in retro-technology goods. The labor-market clearing conditions decouple by technology:

$$Q_n \cdot \ell_A^{\text{firm}} = L_A, \quad (1 - Q_n) \cdot \ell_B^{\text{firm}} = L_B, \quad (11)$$

where L_A, L_B are the exogenous endowments of factors A, B . Since ℓ_s^{firm} is determined by the firm's cost-minimizing labor demand at given prices, Q_n is pinned by the relative factor endowments L_A/L_B and is no longer a free firm choice. The constrained-efficient planner faces the same constraint, so Q_n^* equals Q_n^{CE} trivially. The wedge is zero.

Case (iv): differentiated inputs under (B). Suppose both technology types use both factors A and B but in different intensities (e.g., new-tech is A -intensive, retro is B -intensive). Factor-market clearing gives equilibrium factor prices $w_A(Q_n), w_B(Q_n)$ with $w'_A(Q_n) > 0$ and $w'_B(Q_n) < 0$, so the firm's marginal cost of producing new tech rises in Q_n .

The firm's first-order condition becomes

$$\Phi'(e_s) = p_e(s, e_s) \left[\Delta\pi + q \Delta J - (c_n w'_A(Q_n) - c_r w'_B(Q_n)) \right], \quad (12)$$

where the bracketed correction is the supply-side internalization through factor prices. Define the firm's effective gradient (the bracket on the right-hand side of (12) divided by λ) as

$$g_f(Q_n) \equiv (1/\lambda) [\Delta\pi + q \Delta J] - (1/\lambda) (c_n w'_A(Q_n) - c_r w'_B(Q_n)), \quad (13)$$

and define the planner's welfare gradient under utilitarian welfare weights $\omega_i = 1$ (equal across consumer types $i \in \{Y, O\}$) as

$$g_p(Q_n) \equiv \frac{1}{\lambda} \frac{dW}{dQ_n} = \sum_i \frac{1}{\lambda} \frac{\partial V_i}{\partial P_n} \frac{\partial P_n}{\partial Q_n} + \sum_i \frac{1}{\lambda} \frac{\partial V_i}{\partial P_r} \frac{\partial P_r}{\partial Q_n} + \frac{1}{\lambda} \frac{\partial V_i}{\partial m_i} \frac{\partial m_i}{\partial Q_n}. \quad (14)$$

Apply Roy's identity type-by-type: $\partial V_i / \partial P_s = -\lambda_i c_{i,s}$, where λ_i is type i 's marginal utility of income and $c_{i,s}$ is type i 's consumption of good s . Substituting (2) and aggregating,

$$g_p(Q_n) = - \sum_i \frac{\lambda_i}{\lambda} \frac{c_{i,n} P_n}{(\varepsilon - 1) Q_n} + \sum_i \frac{\lambda_i}{\lambda} \frac{c_{i,r} P_r}{(\varepsilon - 1) (1 - Q_n)} + \frac{1}{\lambda} \sum_i \lambda_i \frac{\partial m_i}{\partial Q_n}. \quad (15)$$

The firm-side gradient (13), by contrast, weights consumption by aggregate consumption shares $C_s / \sum_i c_{i,s}$ rather than by the type-specific marginal utilities λ_i . Define the wedge between aggregate consumption and welfare-weighted consumption as

$$\Delta_s(Q_n) \equiv \sum_i (\lambda_i / \lambda - 1) c_{i,s} \quad (s \in \{n, r\}). \quad (16)$$

Then

$$g_p(Q_n) - g_f(Q_n) = -\frac{\Delta_n(Q_n)P_n}{(\varepsilon-1)Q_n} + \frac{\Delta_r(Q_n)P_r}{(\varepsilon-1)(1-Q_n)} + \frac{1}{\lambda} (c_n w'_A(Q_n) - c_r w'_B(Q_n)), \quad (17)$$

where the third term on the right-hand side is the factor-price correction *added back* (since the firm has already internalized it). Under (B), $\lambda_Y \neq \lambda_O$ generically (the young's marginal utility of income differs from the old's), so Δ_n, Δ_r do not vanish, and the first two terms in (17) are non-zero. The factor-price correction (the third term) only affects the supply side; it does not annihilate the type-heterogeneity terms. Hence $g_p(Q_n) \neq g_f(Q_n)$ generically, and $Q_n^{CE} \neq Q_n^*$.

If, in addition to differentiated inputs, demand is also homogenized ((B) fails), then $\lambda_i = \lambda$ for all i , so $\Delta_s = 0$ identically. The first two terms in (17) vanish, and the factor-price correction (12) is exactly the residual that closes the gap with the planner. The proportionality argument of Case (ii) carries through, and the wedge vanishes. ■